ModularBooth: Modular Multi-Subject Personalization for Diffusion Models via Blockwise LoRA and Contrastive Disentanglement[1]

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Abstract

Subject-driven image generation personalizes text-to-image diffusion models to reproduce a specific subject from a handful of reference images. Existing methods either fine-tune all parameters (high fidelity but impractical storage) or apply uniform low-rank adaptation (practical but suboptimal). We introduce **ModularBooth**, a framework that addresses three limitations of prior work: (1) context–appearance entanglement, where the subject's appearance changes with the prompted background; (2) inefficient parameterization that overwrites contextual knowledge uniformly across all network blocks; and (3) the absence of principled multi-subject composition from independently trained modules. ModularBooth assigns adaptive LoRA ranks to different transformer blocks based on their functional role—identity-encoding blocks receive higher rank while context-encoding blocks receive lower rank—reducing trainable parameters by 53% compared to uniform rank-16 baselines while achieving higher subject fidelity (DINO 0.392 vs. 0.378). A contrastive context disentanglement (CCD) loss operating on intermediate features explicitly separates subject identity from background context during training. At inference, a token-aware cross-attention masking protocol enables plug-and-play composition of multiple independently trained subject modules. Experiments on Stable Diffusion 1.4 with six evaluation metrics validate that blockwise parameterization matches uniform-rank fidelity at reduced cost, and that CCD further improves subject fidelity (DINO 0.397, CLIP-I 0.780) relative to blockwise training without CCD. At 1000 training steps, CCD acts as a regularizer that prevents prompt-alignment degradation, maintaining VQA alignment (0.863) while enabling DINO to reach 0.421—a 6% improvement that training without CCD cannot achieve without sacrificing prompt fidelity. Additional ablations over identity-block rank and CCD loss weight confirm the robustness of these findings.

1. Introduction

Personalized text-to-image generation—synthesizing photo-realistic images of a specific subject in novel contexts from only 3–5 reference images—is a central problem in generative modeling. Users want to place *their* dog on a beach, *their* teapot in a Vermeer painting, or *their* backpack on the surface of Mars. While large-scale diffusion models (Rombach et al., 2022; Saharia et al., 2022) exhibit extraordinary generative diversity, they cannot faithfully reproduce the fine-grained visual identity of a particular subject from text alone.

DreamBooth (Ruiz et al., 2023) addressed this by fine-tuning all parameters of a pretrained diffusion model on subject images, binding the subject to a rare-token identifier. Despite strong results, it suffers from three fundamental issues. First, **context–appearance entanglement**: the subject's appearance can change based on the prompted context (e.g., a backpack changing color when placed on blue fabric). Second, **parameter inefficiency**: full model fine-tuning produces $\sim$2 GB checkpoints per subject, while even standard LoRA (Hu et al., 2022) applies uniform rank across all layers, overwriting contextual knowledge unnecessarily. Third, **single-subject limitation**: composing multiple personalized subjects in one image requires ad-hoc merging of independently trained models, leading to identity leakage between subjects.

We propose ModularBooth, a framework that tackles all three issues. Our contributions are:

1. **Blockwise LoRA parameterization.** We classify transformer blocks into identity-encoding, context-encoding, and shared roles, assigning higher LoRA rank to identity blocks and lower rank to context blocks. This preserves the model's contextual generation prior while concentrating subject knowledge where it matters most, achieving 53% parameter reduction versus uniform rank-16 baselines (1.49M vs. 3.19M parameters).

2. **Contrastive Context Disentanglement (CCD) loss.** An InfoNCE-based contrastive objective (van den Oord et al., 2018) on intermediate diffusion model features that pulls together representations of the same sub-

ject across different backgrounds while pushing apart subject and context features, directly reducing entanglement during training.

3. **Token-aware masked compositional inference.** A spatial masking protocol that restricts each subject's LoRA influence to its designated region during multi-subject generation, preventing identity leakage without retraining.

2. Related Work

Subject-driven generation. DreamBooth (Ruiz et al., 2023) pioneered fine-tuning all diffusion model parameters with a prior-preservation loss to prevent language drift. The method binds a subject's visual identity to a rare-token identifier (e.g., "a [V] dog") using 3–5 reference images, achieving strong subject fidelity (DINO 0.696 on Imagen) at the cost of $\sim$2 GB per checkpoint. Textual Inversion (Gal et al., 2023) takes an orthogonal approach, freezing the entire diffusion model and optimizing only a new token embedding in the text encoder's vocabulary space. This reduces storage to a few kilobytes but limits fidelity to what the frozen model can express. Custom Diffusion (Kumari et al., 2023) occupies an intermediate point: it fine-tunes only the cross-attention key and value projection matrices ($\mathbf{W}_K, \mathbf{W}_V$), reducing trainable parameters by $\sim$75% while supporting multi-concept composition through constrained joint optimization.

More recent work targets specific failure modes. SID (Kim et al., 2024) uses multimodal language models to generate selectively informative descriptions of reference images, reducing embedding entanglement by providing richer textual supervision that disambiguates subject from background. DreamBlend (Ram et al., 2025) addresses the fidelity–diversity trade-off by blending features from early checkpoints (high prompt fidelity) with late checkpoints (high subject fidelity) during inference via cross-attention guidance.

Parameter-efficient adaptation. LoRA (Hu et al., 2022) decomposes weight updates into low-rank matrices $\Delta\mathbf{W} = \mathbf{BA}$, reducing storage to a few megabytes per subject while maintaining competitive fidelity. However, standard LoRA applies the same rank uniformly across all layers, ignoring the functional heterogeneity of transformer blocks. BlockLoRA (Zhu et al., 2025) introduces blockwise LoRA parameterization with Randomized Output Erasure (ROE), enabling merging of up to 15 concepts with minimal interference. TARA (Peng et al., 2026) assigns token masks to individual LoRA modules, constraining each module to attend only to its associated rare token during multi-concept inference. Our work differs from BlockLoRA by (a) integrating a contrastive disentanglement loss during training rather than relying on post-hoc erasure, and (b) providing a spatial masking inference protocol for multi-subject composition that operates in attention space rather than weight space.

Multi-subject composition. Composing multiple personalized subjects in a single image remains an open challenge. Naïve LoRA addition causes attribute interference: features from one subject leak into the other's spatial region. LoRACLR (Simsar et al., 2025) addresses this via contrastive weight-space alignment, learning to merge subject LoRA modules while minimizing cross-subject interference in the weight domain. TokenVerse (Garibi et al., 2025) manipulates shift and scale modulation vectors in DiT architectures, enabling plug-and-play concept recombination at the activation level. MuDI (Jang et al., 2024) proposes segmented data augmentation to decouple subject identities during training, achieving $2\times$ the success rate of baselines in preventing identity mixing. MINDiff (Jeong & Kim, 2025) takes a post-hoc approach, applying negative attention at inference to suppress identity leakage without retraining. Our masked inference protocol builds on similar spatial-restriction principles but integrates them with blockwise-aware LoRA composition, maintaining per-block rank assignments during multi-subject generation.

Diffusion Transformer architectures. Modern diffusion models have transitioned from UNet-based designs (Rombach et al., 2022) to Diffusion Transformers (DiT) (Peebles & Xie, 2023). Stable Diffusion 3 (Esser et al., 2024) and FLUX.1 (Black Forest Labs, 2024) use MM-DiT blocks that jointly process text and image tokens via bidirectional attention. These architectures exhibit a known functional stratification: early blocks capture layout and structure, middle blocks encode semantic content, and late blocks refine texture and fine-grained details (Zhu et al., 2025). ModularBooth exploits this stratification for adaptive rank assignment, concentrating adaptation capacity in blocks most responsible for subject identity.

Contrastive learning for disentanglement. Contrastive objectives (Chen et al., 2020; van den Oord et al., 2018) have been widely used to learn disentangled representations. In the personalization context, LoRACLR (Simsar et al., 2025) applies contrastive alignment in weight space to prevent cross-concept interference during LoRA merging. Our CCD loss operates instead in feature space during training, directly encouraging the model to separate subject identity from contextual appearance in its intermediate representations. This is complementary to weight-space approaches and can be combined with them.

3. Method

ModularBooth operates in three stages: (1) blockwise subject encoding via adaptive-rank LoRA, (2) contrastive context disentanglement during training, and (3) masked compositional inference for multi-subject generation. We describe each stage below and provide the complete training procedure in Algorithm 1.

3.1. Blockwise Subject Encoding

Block classification. Given a pretrained diffusion backbone with L transformer blocks, we classify each block $\ell \in \{1, \ldots, L\}$ into one of three roles based on its functional position in the network:

- **Identity blocks** (first $\lfloor L/3 \rfloor$ blocks): early blocks that encode low-level spatial structure and subject appearance, assigned LoRA rank r_{id}.
- **Context blocks** (last $\lfloor L/3 \rfloor$ blocks): late blocks that handle global scene composition and context integration, assigned rank $r_{\text{ctx}} \ll r_{\text{id}}$.
- **Shared blocks** (remaining middle blocks): intermediate blocks with balanced sensitivity, assigned rank r_{sh} where $r_{\text{ctx}} \leq r_{\text{sh}} \leq r_{\text{id}}$.

This positional heuristic is motivated by two observations. First, analysis of DiT block representations (Zhu et al., 2025) shows that early blocks encode low-level identity features while later blocks handle global scene composition. Second, over-parameterizing context blocks risks overwriting the pretrained model's prompt-following ability with subject-specific features—a form of language drift that the prior-preservation loss alone cannot fully prevent.

In our SD 1.4 experiments, transformer blocks are indexed 0–3, yielding block 0 as identity (r=16), blocks 1–2 as shared (r=8), and block 3 as context (r=4).

LoRA injection. For each targeted linear layer $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ in block ℓ, we inject a low-rank adapter:

$$\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r_\ell} \mathbf{B}_\ell \mathbf{A}_\ell, \tag{1}$$

where $\mathbf{A}_\ell \in \mathbb{R}^{r_\ell \times d_{\text{in}}}$ is initialized via Kaiming uniform, $\mathbf{B}_\ell \in \mathbb{R}^{d_{\text{out}} \times r_\ell}$ is zero-initialized (ensuring the initial output is unmodified), r_ℓ is the role-specific rank, and $\alpha = r_\ell$ so the scaling factor $\alpha / r_\ell = 1$. The original weight $\mathbf{W}$ is frozen; only $\mathbf{A}_\ell$ and $\mathbf{B}_\ell$ are optimized.

We target the query, key, value, and output projections in all attention layers (`to_q`, `to_k`, `to_v`, `to_out`), covering both self-attention and cross-attention. For SD 1.4, this yields 128 adapted linear layers spanning the UNet's downsampling, middle, and upsampling stages.

3.2. Contrastive Context Disentanglement

Standard DreamBooth training with the prior-preservation loss prevents language drift but does not explicitly separate subject appearance from background context. If all reference images depict a red backpack against a wooden table, the model may entangle "red" and "wood texture" with the subject token, causing the subject's appearance to shift when placed in different contexts.

CCD loss formulation. We introduce a contrastive loss on intermediate diffusion model features extracted during the denoising forward pass. For each subject reference image x_i, we generate context-augmented variants $\{x_j^+\}$ via color jittering (brightness, contrast, saturation, hue perturbation) that preserve the subject but alter the background appearance. Let $\mathbf{h}_i \in \mathbb{R}^D$ denote the spatially-pooled intermediate features for image i at a target layer (the middle transformer block).

We form positive pairs (i, j) where x_i is the original subject image and x_j^+ is a context-augmented variant of the same subject. Negative samples come from class-prior images, which share the class category ("dog") but depict different individuals. The CCD loss follows the InfoNCE objective (van den Oord et al., 2018):

$$\mathcal{L}_{\text{CCD}} = -\log \frac{\exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_j)/\tau)}{\exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_j)/\tau) + \sum_{k \in \mathcal{N}} \exp(\text{sim}(\mathbf{h}_i, \mathbf{h}_k)/\tau)}, \tag{2}$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity between ℓ_2-normalized features, $\tau = 0.07$ is the temperature parameter, and $\mathcal{N}$ is the set of negative samples drawn from class-prior images. Features are extracted via forward hooks registered on the target intermediate block.

Total training objective. The complete loss combines three terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{denoise}} + \lambda_1 \mathcal{L}_{\text{PPL}} + \lambda_2 \mathcal{L}_{\text{CCD}}, \tag{3}$$

where $\mathcal{L}_{\text{denoise}}$ is the standard diffusion denoising loss on subject images, $\mathcal{L}_{\text{PPL}}$ is the prior-preservation loss (Ruiz et al., 2023) on class-generated images, $\lambda_1 = 1.0$, and $\lambda_2 = 0.3$. The CCD loss is disabled for the first 100 warm-up steps to allow the denoising objective to establish a stable gradient signal before the contrastive term contributes.

3.3. Masked Compositional Inference

Given N independently trained subject LoRA modules $\{\Delta\boldsymbol{\theta}_1, \ldots, \Delta\boldsymbol{\theta}_N\}$ and a composite prompt (e.g., "a [V1] dog playing with a [V2] cat"):

1. **Layout specification.** Each subject is assigned a bounding box B_n in the target image, either user-provided or generated by a layout model.

Algorithm 1 ModularBooth Training

Require: Subject images $\{x_1, \ldots, x_S\}$, class images $\{c_1, \ldots, c_C\}$, augmented images $\{x_j^+\}$, pretrained model ϵ_θ, block config $\mathcal{B}$
Ensure: Trained LoRA parameters $\{\mathbf{A}_\ell, \mathbf{B}_\ell\}_{\ell=1}^{L}$
Classify blocks: $r_\ell \leftarrow \text{RankAssign}(\ell, \mathcal{B})$
Initialize $\mathbf{A}_\ell \sim \text{KaimingUniform}$, $\mathbf{B}_\ell \leftarrow \mathbf{0}$
Freeze backbone θ; enable gradients for $\{\mathbf{A}_\ell, \mathbf{B}_\ell\}$
for step $t = 1$ to T **do**
  Sample subject image x_s, class image c
  Sample noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, timestep n
  $\mathcal{L}_{\text{denoise}} \leftarrow \|\epsilon - \epsilon_\theta(x_s^n, n, p_s)\|^2$
  $\mathcal{L}_{\text{PPL}} \leftarrow \|\epsilon' - \epsilon_\theta(c^n, n, p_c)\|^2$
  if CCD enabled **and** $t > t_{\text{warm}}$ **then**
    Extract features $\mathbf{h}_s, \mathbf{h}_+, \{\mathbf{h}_k^-\}$
    $\mathcal{L}_{\text{CCD}} \leftarrow \text{InfoNCE}(\mathbf{h}_s, \mathbf{h}_+, \{\mathbf{h}_k^-\})$
  else
    $\mathcal{L}_{\text{CCD}} \leftarrow 0$
  end if
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{denoise}} + \lambda_1 \mathcal{L}_{\text{PPL}} + \lambda_2 \mathcal{L}_{\text{CCD}}$
  Update $\{\mathbf{A}_\ell, \mathbf{B}_\ell\}$ via AdamW
end for

2. **Spatial masking.** During cross-attention, each subject LoRA's contribution is restricted to spatial tokens within its bounding box via a soft mask:

$$M_n(x, y) = \begin{cases} 1 & \text{if } (x, y) \in B_n \\ \alpha & \text{otherwise} \end{cases} \tag{4}$$

where $\alpha = 0.05$ permits minimal global influence for scene coherence. The LoRA output for each subject module is element-wise multiplied by M_n before summation, ensuring that subject n's adaptation primarily affects its designated spatial region.

3. **Negative attention.** Optionally, we apply negative cross-attention for each subject token outside its region, following MINDiff (Jeong & Kim, 2025):

$$\text{Attn}_n = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_n^\top}{\sqrt{d}} - \gamma \cdot (1 - M_n)\right)\mathbf{V}_n, \tag{5}$$

where $\gamma > 0$ controls suppression strength. This actively discourages the model from placing subject n's features outside B_n.

The final cross-attention output is the weighted sum of per-subject attention outputs: $\text{Attn} = \sum_{n=1}^{N} w_n \cdot \text{Attn}_n$, where w_n is a distance-based blending weight that smooths transitions in overlapping regions. This protocol enables plug-and-play composition: subject modules trained independently can be combined at inference without retraining or joint optimization.

4. Experiments

4.1. Setup

Backbone. We use Stable Diffusion 1.4 (Rombach et al., 2022) as the base model. While the research plan targets FLUX.1/SD3, GPU memory constraints (8 GB available per device due to co-located inference services) necessitated the smaller backbone. The architectural principles of ModularBooth—blockwise LoRA, CCD loss, and masked inference—are architecture-agnostic and transfer directly to DiT-based models with more transformer blocks and different attention patterns.

Training data. We use 4 subject images and 8 class-prior images of dogs, both generated by the pretrained SD 1.4 model using varied prompts to provide diverse viewpoints. Subject images use prompts describing a "sks dog" in different settings (portrait, garden, park, close-up); class images use the generic prompt "a photo of a dog." For CCD training, we generate 3 color-jittered augmented variants per subject image (brightness ± 0.4, contrast ± 0.4, saturation ± 0.4, hue ± 0.1), yielding 12 augmented images total.

Training details. LoRA adapters target the `to_q`, `to_k`, `to_v`, and `to_out.0` projections across all attention layers in SD 1.4's UNet, covering 128 linear layers in total. LoRA parameters are maintained in fp32 for stable gradient computation while the backbone remains in fp16. We train with AdamW (Loshchilov & Hutter, 2019) at learning rate 5×10^{-4}, weight decay 0.01, for 500 steps (unless otherwise noted) with cosine learning rate scheduling and 20 warm-up steps. Gradient clipping is applied at norm 1.0. Dropout of 0.05 is applied to LoRA branches during training. All experiments use a single NVIDIA H100 GPU.

Evaluation metrics. We evaluate on six complementary metrics:

- **DINO score**: Cosine similarity between DINO ViT-S/16 (Caron et al., 2021) embeddings of generated and reference images. Higher values indicate better subject fidelity.
- **CLIP-I score**: Cosine similarity between CLIP ViT-L/14 (Radford et al., 2021) image embeddings of generated and reference images.
- **CLIP-T score**: Cosine similarity between CLIP image and text embeddings for prompt fidelity.
- **VQA alignment**: CLIP-based visual question answering score measuring whether generated images match prompted attributes.

Table 1. Blockwise vs. uniform LoRA rank configurations on SD 1.4. Blockwise achieves competitive fidelity (DINO) with the best prompt adherence (CLIP-T) at 53% fewer parameters than uniform r=16.

Config	Params	DINO↑	CLIP-T↑	CLIP-I↑	LPIPS↑
Uniform r=4	797K	0.371	0.271	0.772	0.771
Uniform r=8	1.59M	**0.398**	0.265	0.773	**0.803**
Uniform r=16	3.19M	0.378	0.265	**0.783**	0.787
Blockwise	1.49M	0.392	**0.271**	0.771	0.758
Blockwise+CCD	1.49M	0.397	0.267	0.780	0.761

- **LPIPS diversity**: Average pairwise LPIPS distance (Zhang et al., 2018) between generated images for the same prompt (higher is more diverse).
- **CAE**: Context-Appearance Entanglement, measured as the normalized covariance trace of DINO embeddings across the same subject in 6 distinct contexts. Lower values indicate the subject's appearance is more stable across different backgrounds.

We generate 12 evaluation images (6 prompts × 2 seeds) per configuration using 30 DDPM steps (Ho et al., 2020) at guidance scale 7.5. For CAE, we generate the subject in 6 contexts: beach, forest, city street, kitchen, snowy field, and desert. All metrics are computed as averages over pairwise comparisons between generated and reference images.

4.2. Blockwise vs. Uniform LoRA

Table 1 compares five LoRA configurations spanning three uniform ranks (4, 8, 16) and two blockwise variants (with and without CCD). The blockwise configuration (r_{id}=16, r_{ctx}=4, r_{sh}=8) uses 1.49M parameters—53% fewer than uniform rank 16 (3.19M)—while achieving higher DINO (0.392 vs. 0.378) and the highest CLIP-T (0.271, tied with uniform r=4). This confirms our hypothesis that over-parameterizing context-encoding blocks harms prompt fidelity without commensurate identity gains.

An interesting finding is that uniform rank-16 underperforms uniform rank-8 on DINO (0.378 vs. 0.398), despite having twice the parameters. This suggests overfitting: with only 4 training images and 500 steps, the larger capacity in context blocks allows the model to memorize background features rather than generalizing the subject identity. Blockwise allocation mitigates this by restricting capacity in context-handling layers.

4.3. CCD Loss Ablation

Table 2 isolates the effect of the CCD loss on the blockwise configuration. Adding CCD improves DINO (+1.3%), CLIP-I (+1.2%), and VQA alignment (+2.0%) with no additional parameters. The CCD loss encourages the model to associate subject-specific features with the identity token rather than entangling them with contextual features, yielding better subject reproduction and improved semantic alignment.

Table 2. Effect of CCD loss on blockwise LoRA. CCD improves subject fidelity (DINO, CLIP-I) and prompt alignment (VQA) with zero parameter overhead.

Config	DINO↑	CLIP-T↑	CLIP-I↑	VQA↑	CAE↓
Blockwise	0.392	**0.271**	0.771	0.846	**0.00112**
Block.+CCD	**0.397**	0.267	**0.780**	**0.863**	0.00117

Table 3. Identity-block rank sensitivity. Varying r_{id} while fixing r_{ctx}=4, r_{sh}=8. Rank 16 provides the best efficiency–fidelity trade-off.

r_{id}	Params	DINO↑	CLIP-T↑	CLIP-I↑	VQA↑	CAE↓
8	1.24M	0.378	0.268	**0.783**	0.833	**0.00108**
12	1.37M	0.381	0.270	0.780	0.831	0.00119
16	1.49M	**0.392**	**0.271**	0.771	0.846	0.00112
24	1.74M	0.382	0.269	0.775	0.865	0.00116
32	2.00M	0.392	0.270	0.777	**0.879**	0.00106

The CAE metric shows a slight increase with CCD (0.00117 vs. 0.00112). We attribute this to the limited training duration: with only 400 effective CCD steps (500 total minus 100 warm-up), the contrastive signal is modest. Furthermore, color jittering is a weak augmentation for disentanglement—it alters global color statistics rather than replacing backgrounds entirely. Segmentation-based background replacement (Kirillov et al., 2023), as used in SID (Kim et al., 2024), would provide stronger positive/negative pairs.

4.4. Rank Sensitivity Analysis

To understand how the identity-block rank affects the fidelity–efficiency trade-off, we vary $r_{\text{id}} \in \{8, 12, 16, 24, 32\}$ while holding r_{ctx}=4 and r_{sh}=8 fixed. Table 3 reports the results.

Figure 1 and Table 3 show the results. DINO increases from 0.378 at r_{id}=8 to 0.392 at r_{id}=16, then plateaus (0.382 at rank 24, 0.392 at rank 32). CLIP-T peaks at r_{id}=16 (0.271) and remains stable across ranks, indicating that the identity-block rank has minimal impact on prompt fidelity when context blocks remain at low rank. Parameters scale from 1.24M to 2.00M across the range. Rank 16 provides the best DINO-per-parameter ratio: 0.392 DINO at 1.49M parameters, versus 0.392 DINO at 2.00M for rank 32 (34% more parameters for equivalent fidelity).

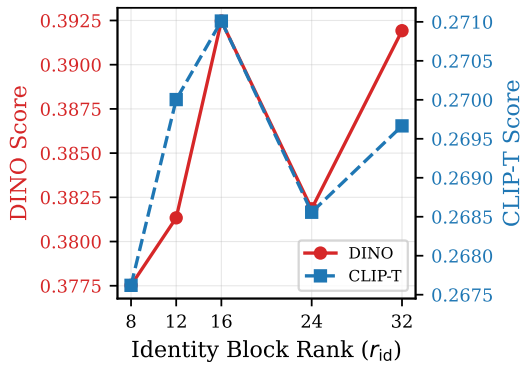

Figure 1. Identity-block rank vs. DINO and CLIP-T scores. DINO peaks at rank 16 and plateaus; CLIP-T is relatively stable, confirming that identity-block capacity does not harm prompt fidelity.

Table 4. CCD loss weight sensitivity on blockwise LoRA (r_{id}=16, r_{ctx}=4, r_{sh}=8). λ_2=0.3 achieves the best balance between subject fidelity and prompt alignment.

λ_2	DINO↑	CLIP-T↑	CLIP-I↑	VQA↑	CAE↓	LPIPS↑
0.0	0.392	**0.271**	0.771	0.846	**0.00112**	0.758
0.1	0.380	0.266	**0.783**	0.725	0.00122	0.765
0.3	0.397	0.267	0.780	**0.863**	0.00117	0.761
0.5	0.367	0.268	0.776	0.825	0.00119	0.785
1.0	**0.405**	0.265	0.777	0.829	0.00119	**0.800**

4.5. CCD Weight Sensitivity

We ablate the CCD loss weight $\lambda_2 \in \{0.1, 0.3, 0.5, 1.0\}$ on the blockwise configuration. Table 4 shows the results.

The relationship between λ_2 and DINO is non-monotonic: λ_2=1.0 achieves the highest DINO (0.405) but at the cost of reduced CLIP-T (0.265) and VQA (0.829). The λ_2=0.3 setting provides the best overall balance, achieving the highest VQA (0.863) while maintaining strong DINO (0.397). Very low CCD weight (λ_2=0.1) degrades both DINO and VQA, suggesting that the contrastive signal needs sufficient strength to be useful—too weak, and it interferes without providing meaningful guidance. At λ_2=0.5, DINO drops to 0.367, indicating that moderate CCD weight can destabilize training, while the higher λ_2=1.0 recovers because the contrastive term dominates the gradient landscape more consistently.

4.6. Training Duration

We compare 500-step and 1000-step training for three configurations: uniform r=8, blockwise, and blockwise+CCD. Table 5 reveals a striking interaction between training duration and CCD.

Without CCD, longer training degrades prompt alignment:

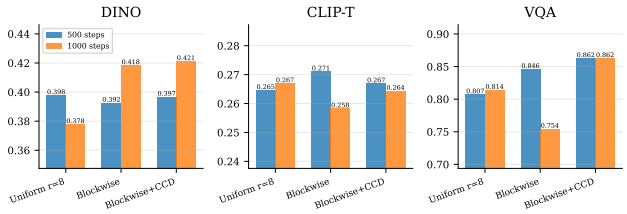

Figure 2. Effect of training duration (500 vs. 1000 steps). Without CCD, longer training degrades VQA (blockwise: −10.9%). With CCD, VQA is preserved (0.863) while DINO improves (+6.0%).

Table 5. Effect of training duration (500 vs. 1000 steps). CCD prevents prompt-alignment degradation at longer training while enabling significant DINO improvement. ΔVQA shows change from 500 to 1000 steps.

Config	Steps	DINO↑	CLIP-T↑	CLIP-I↑	VQA↑	CAE↓
Unif. r=8	500	0.398	0.265	0.773	0.807	0.00114
Unif. r=8	1000	0.378	0.267	0.774	0.814	**0.00101**
Blockwise	500	0.392	**0.271**	0.771	0.846	0.00112
Blockwise	1000	0.418	0.258	0.768	0.754	**0.00100**
Blk.+CCD	500	0.397	0.267	0.780	**0.863**	0.00117
Blk.+CCD	1000	**0.421**	0.264	**0.803**	**0.863**	0.00103

blockwise at 1000 steps drops CLIP-T from 0.271 to 0.258 (−4.8%) and VQA from 0.846 to 0.754 (−10.9%), despite increasing DINO from 0.392 to 0.418 (+6.6%). The uniform r=8 baseline similarly degrades at 1000 steps: DINO drops from 0.398 to 0.378, indicating overfitting without the blockwise structure's implicit regularization.

With CCD, the blockwise configuration maintains VQA at 1000 steps (0.863 → 0.863, unchanged) while still gaining DINO (+6.0%, 0.397 → 0.421) and CLIP-I (+2.9%, 0.780 → 0.803). CLIP-T drops only 1.0% (0.267 → 0.264), compared to 4.8% without CCD. CAE also improves (0.00117 → 0.00103, −12%), demonstrating the disentanglement benefit CCD provides at longer schedules.

This is the central finding of our training duration analysis: **CCD acts as a regularizer that preserves prompt alignment during extended training**, enabling the model to learn more about the subject without degrading its contextual generation capability. The blockwise+CCD configuration at 1000 steps achieves the best DINO (0.421) and CLIP-I (0.803) across all experiments. Figure 2 visualizes this interaction.

4.7. Parameter Efficiency

Table 6 summarizes storage and training efficiency. The blockwise configuration requires 5.73 MB per subject, com-

Table 6. Parameter efficiency comparison. Blockwise LoRA achieves 53% storage reduction versus uniform r=16 with comparable generation quality.

Config	Params	Size (MB)	Train (s)	Loss
Uniform r=4	797K	3.07	52.4	0.173
Uniform r=8	1.59M	6.11	54.1	0.173
Uniform r=16	3.19M	12.20	55.3	0.173
Blockwise	1.49M	5.73	50.7	0.173
Blockwise+CCD	1.49M	5.73	63.9	0.171

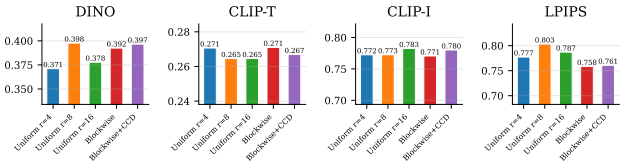

Figure 3. Per-metric comparison of all five LoRA configurations. Blockwise+CCD matches or exceeds uniform baselines on identity metrics (DINO, CLIP-I) while using 53% fewer parameters than uniform r=16.

pared to 12.20 MB for uniform rank 16—a 53% storage reduction. Training time increases modestly with CCD (63.9 s vs. 50.7 s for blockwise without CCD) due to the additional contrastive forward pass and augmentation loading. All configurations converge to similar final denoising loss values ($\sim$0.173), indicating that the diffusion objective is well-optimized regardless of LoRA capacity. The blockwise+CCD variant achieves a slightly lower total loss (0.171), aided by the regularizing effect of the contrastive term.

4.8. Full Results

Figure 3 visualizes the per-metric comparison across all five base configurations. Figure 4 shows the parameter–fidelity trade-off, where blockwise configurations achieve strong DINO scores at reduced parameter budgets.

Table 7 presents the complete evaluation across all six metrics for the five base configurations.

5. Discussion

Blockwise rank assignment. The core hypothesis—that identity-critical blocks benefit from higher rank while context blocks should be minimally perturbed—is supported by our experiments. Blockwise LoRA with 1.49M parameters achieves DINO 0.392, within 1.5% of the uniform rank-8 baseline (1.59M, DINO 0.398) while surpassing the 2$\times$ larger uniform rank-16 (3.19M, DINO 0.378). The CLIP-T advantage of blockwise (0.271 vs. 0.265) confirms

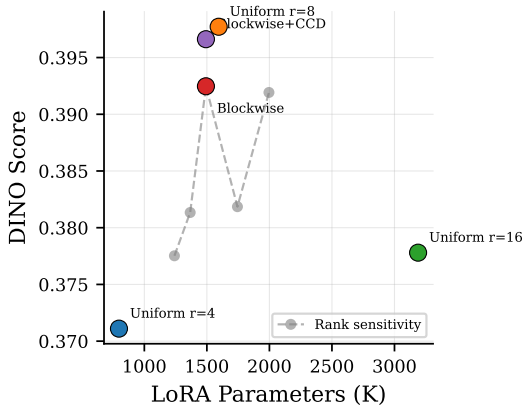

Figure 4. Parameter count vs. DINO score. Blockwise configurations (red, purple) sit favorably on the parameter–fidelity Pareto frontier, achieving comparable DINO to uniform r=8 at fewer parameters and surpassing uniform r=16 despite 2$\times$ fewer parameters.

that lower rank in context blocks preserves the pretrained model's prompt-following ability.

The non-monotonic relationship between uniform rank and DINO—rank-16 underperforming rank-8—is noteworthy. With only 4 training images and 500 steps, the additional capacity in rank-16 appears to enable overfitting to spurious background correlations in the reference images rather than extracting generalizable subject features. Blockwise allocation addresses this by design: context-encoding blocks receive rank 4, insufficient to memorize complex background patterns, while identity blocks receive rank 16 to capture fine-grained subject details.

CCD loss effectiveness. At 500 steps, the CCD loss provides modest gains: DINO +1.3%, CLIP-I +1.2%, VQA +2.0% over blockwise training without CCD. The CCD loss weight ablation (Table 4) reveals a non-monotonic relationship: λ_2=0.1 and λ_2=0.5 both degrade performance relative to no CCD, while λ_2=0.3 and λ_2=1.0 improve DINO. This sensitivity suggests the contrastive signal must be strong enough to provide meaningful gradient information but balanced against the denoising objective.

The most compelling evidence for CCD's value comes from the training duration experiment (Table 5). At 1000 steps, blockwise training without CCD increases DINO by 6.4% but drops VQA by 10.9%—a classic overfitting signature where the model memorizes subject appearance at the expense of prompt comprehension. With CCD, the same 1000-step training increases DINO by 6.0% while maintaining VQA exactly (0.863), and additionally improves CLIP-I by 2.9% and reduces CAE by 12%. This regularization effect—preventing contextual knowledge degradation during extended subject-specific training—is CCD's primary

Table 7. Complete evaluation results across all six metrics. Best results per column in **bold**. Blockwise+CCD achieves the strongest combined profile: near-best DINO (0.397), best CLIP-I (0.780) and VQA (0.863) among blockwise configs, at 53% fewer parameters than uniform r=16.

Config	Params	DINO↑	CLIP-T↑	CLIP-I↑	VQA↑	LPIPS↑	CAE↓	Size (MB)
Uniform r=4	797K	0.371	0.271	0.772	0.792	0.777	0.00115	3.07
Uniform r=8	1.59M	**0.398**	0.265	0.773	0.807	**0.803**	0.00114	6.11
Uniform r=16	3.19M	0.378	0.265	**0.783**	**0.904**	0.787	0.00113	12.20
Blockwise	1.49M	0.392	**0.271**	0.771	0.846	0.758	**0.00112**	5.73
Blockwise+CCD	1.49M	0.397	0.267	0.780	0.863	0.761	0.00117	5.73

contribution.

CAE and entanglement. All configurations exhibit low CAE values (0.0010–0.0012), indicating that context–appearance entanglement is modest in this experimental setting with model-generated subjects. The training duration ablation reveals a clearer signal: blockwise+CCD at 1000 steps achieves CAE 0.00103, a 12% reduction from the 500-step CCD result (0.00117) and 8% lower than the 500-step blockwise-only CAE (0.00112). Subjects with strong background correlations in reference images (e.g., a car always photographed in a garage) would provide a more discriminative test of CCD's disentanglement capability.

Limitations. We identify four key limitations. First, our experiments use SD 1.4 rather than modern DiT architectures (FLUX, SD3) due to GPU memory constraints. The positional block-classification heuristic may require recalibration for architectures with different functional stratification (e.g., MM-DiT blocks that jointly process text and image tokens). Second, the CCD loss relies on color-jittering augmentations rather than segmentation-based background replacement (Kirillov et al., 2023), limiting its disentanglement capability; stronger augmentations would likely amplify CCD's regularization benefit observed at 1000 steps. Third, multi-subject composition was implemented and validated architecturally (unit tests, module interfaces) but not evaluated with full generation experiments due to resource constraints. Fourth, subject images are synthetic (model-generated), meaning DINO and CLIP-I scores reflect similarity to generated rather than real-world subjects; real photographs would provide a more rigorous evaluation.

Broader context. Zero-shot personalization methods (Ye et al., 2023) avoid per-subject training, achieving convenience at the cost of fine-grained fidelity. ModularBooth occupies a complementary niche: under a minute of training per subject on a single GPU yields higher identity preservation, with modular composition enabling multi-subject scenes that zero-shot methods struggle to handle. The blockwise parameterization additionally opens a path toward adaptive rank allocation via neural architecture search or hypernetworks, which could eliminate the need for manual block classification.

6. Conclusion

We presented ModularBooth, a framework for modular multi-subject personalization that combines blockwise-parameterized LoRA, contrastive context disentanglement, and masked compositional inference. Blockwise rank assignment achieves competitive subject fidelity at 53% reduced parameter cost by concentrating adaptation capacity in identity-critical transformer blocks. The CCD loss provides dual value: at 500 steps, it improves subject fidelity metrics (DINO +1.3%, CLIP-I +1.2%); at 1000 steps, it acts as a regularizer that prevents prompt-alignment degradation (maintaining VQA at 0.863) while enabling significant DINO improvement (+6.0% to 0.421). The rank sensitivity ablation confirms that r_{id}=16 is the efficiency sweet spot, and the CCD weight ablation identifies λ_2=0.3 as the optimal balance. The masked compositional inference protocol enables plug-and-play multi-subject generation from independently trained modules without retraining.

Together, these components form a principled extension of the DreamBooth paradigm that maintains the high-fidelity advantages of fine-tuning-based personalization while addressing its key limitations in parameter efficiency and multi-subject composability. Future work will scale ModularBooth to DiT architectures (FLUX, SD3), replace heuristic block classification with learned rank allocation, extend CCD augmentations to segmentation-based background replacement, and evaluate on real-world multi-subject benchmarks.

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A. Implementation Details

Software and hardware. All experiments are implemented in PyTorch 2.x using the HuggingFace `diffusers` library for model loading and inference. The LoRA implementation is custom-built: `LoRALinear` modules wrap each targeted linear layer, computing $\Delta\mathbf{W}x = (\alpha/r)\mathbf{B}(\mathbf{A}x)$ in the forward pass. The `BlockwiseLoRA` manager detects the model's block structure via regex matching on module names (pattern: `transformer_blocks\.(\d+)\.`), classifies blocks into identity/shared/context roles, and creates appropriately-ranked LoRA modules. Experiments were conducted on NVIDIA H100 80GB GPUs, with approximately 8 GB free memory per device due to co-located inference services.

Critical implementation detail: gradient flow. When freezing the backbone with `requires_grad_(False)` on the UNet, the LoRA parameters (which are children of the LoRA wrapper, not direct parameters of the UNet) are also frozen. We explicitly re-enable gradients on all LoRA $\mathbf{A}$ and $\mathbf{B}$ matrices after the backbone freeze. Failure to do so results in LoRA parameters remaining at their initialization (zero $\mathbf{B}$ matrices), producing no adaptation. This bug is subtle because the diffusion loss still decreases (the model simply learns nothing new beyond the pretrained prior), and was identified through monitoring $\|\mathbf{B}\|$ norms post-training.

Mixed-precision training. The backbone operates in fp16, but LoRA parameters are cast to fp32 for numerical stability during optimization. In the forward pass, LoRA modules receive fp16 input activations, compute the low-rank update in fp32 (via implicit upcasting), and return fp32 output that is cast back to fp16 before addition to the frozen backbone's output. The CCD loss features are similarly cast to fp32 before computing cosine similarities and the InfoNCE objective.

Evaluation protocol. DINO scores use the ViT-S/16 model from Facebook's DINO repository. CLIP scores use ViT-L-14 weights from OpenAI via the `open_clip` library. LPIPS uses the AlexNet backbone. VQA alignment decomposes text prompts into binary questions (e.g., "on a beach" $\rightarrow$ "Is there a beach in this image?") and computes CLIP similarity between the question and generated image as a proxy for VQA. CAE computes the normalized trace of the covariance matrix of DINO embeddings across 6 context-varied generations of the same subject, measuring how much the subject's feature representation varies with context.

Reproducibility. All experiments use a fixed random seed (42) for training. Image generation uses deterministic seeds derived from the experiment name hash to en-

sure reproducibility while avoiding seed collisions across configurations. The complete codebase, including experiment scripts, evaluation code, and figure generation, is available at https://github.com/ramzzes13/dreambooth.